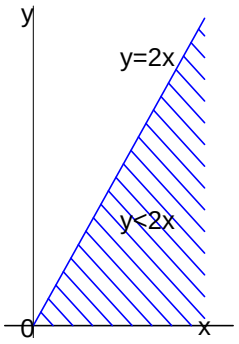
F78PA Probability & Statistics A

Tutorial 9 – Solutions

1. (a) With $X \sim \text{Exp}(1)$ and $Y \sim \text{Exp}(1)$. the corresponding density functions are $f_X(x) = e^{-x}$ for $x \geq 0$ (and $f_X(x) = 0$ for $x < 0$) and $f_Y(y) = e^{-y}$ for $y \geq 0$ (and $f_Y(y) = 0$ for $y < 0$).
Then since X, Y are independent, their joint density function is

$$\begin{aligned} f_{X,Y}(x,y) &= f_X(x) f_Y(y) \qquad \text{(independence)} \\ &= \begin{cases} e^{-(x+y)} & x, y \geq 0, \\ 0 & \text{otherwise.} x \end{cases} \end{aligned}$$

To compute $\mathbb{P}(2X > Y)$, integrate over the region of non-zero density within which this inequality is satisfied — shaded region below.

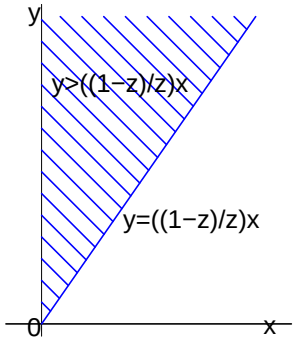


$$\begin{aligned} \mathbb{P}(2X > Y) = \int_{x=0}^{\infty} \int_{y=0}^{2x} f_{X,Y}(x,y)\, dy\, dx &= \int_{x=0}^{\infty} \int_{y=0}^{2x} e^{-x} e^{-y}\, dy\, dx \\ &= \int_{x=0}^{\infty} \left[-e^{-x} e^{-y} \right]_{y=0}^{2x} dx \\ &= \int_{x=0}^{\infty} \left(-e^{-3x} + e^{-x} \right) dx \\ &= \left[\frac{e^{-3x}}{3} - e^{-x} \right]_{x=0}^{\infty} = 0 - \left(\frac{1}{3} - 1 \right) = \frac{2}{3} \end{aligned}$$

(b) With $Z = X/(X+Y)$, since X and Y are both positive then Z is positive, and also $X + Y > X$ so that $Z < 1$. Also, since X, Y can each take any positive value, then $Z = X/(X+Y)$ can take any value between 0 and 1. So $\text{Im} Z = (0, 1)$, as required.
The distribution function $F_Z(z)$ therefore satisfies $F_Z(z) = 0$ for $z \leq 0$ and $F_Z(z) = 1$ for $z \geq 1$.
For $0 < z < 1$,

$$\begin{aligned} F_Z(z) = \mathbb{P}(Z \leq z) &= \mathbb{P}\left(\frac{X}{X+Y} \leq z \right) \\ &= \mathbb{P}\left(X \leq z(X+Y) \right) \\ &= \mathbb{P}\left((1-z)X \leq zY \right) \\ &= \mathbb{P}\left(\left(\frac{1-z}{z} \right) X \leq Y \right) \end{aligned}$$

Need to integrate over shaded region below:



So for $0 < z < 1$,

$$\begin{aligned} F_Z(z) = \mathbb{P}(Z \le z) = \mathbb{P}\left(\left(\frac{1-z}{z}\right) X \le Y\right) &= \int_{x=0}^{\infty} \int_{y=\left(\frac{1-z}{z}\right)x}^{\infty} e^{-x} e^{-y} \, dy \, dx \\ &= \int_{x=0}^{\infty} \left[-e^{-x} e^{-y}\right]_{y=\left(\frac{1-z}{z}\right)x}^{\infty} dx \\ &= \int_{x=0}^{\infty} e^{-x} e^{-\left(\frac{1-z}{z}\right)x} \, dx \\ &= \int_{x=0}^{\infty} e^{-x/z} \, dx \\ &= \left[-z e^{-x/z}\right]_{x=0}^{\infty} \\ &= z \end{aligned}$$

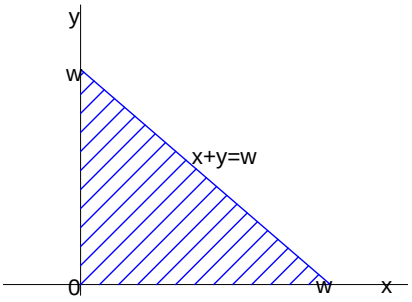
So we have

$$F_Z(z) = \begin{cases} 0 & z < 0, \\ z & 0 \le z < 1, \\ 1 & z \ge 1. \end{cases}$$

This matches the cdf of the uniform distribution on the interval $(0, 1)$, so we have shown that $Z \sim \mathrm{U}(0, 1)$.

(c) With $W = X + Y$, clearly $\mathrm{Im}W = [0, \infty)$, so $F_W(w) = \mathbb{P}(W \le w) = 0$ for $w < 0$.

For $w \ge 0$, $F_W(w) = \mathbb{P}(W \le w) = \mathbb{P}(X + Y \le w)$. To find this probability, integrate over shaded region below:



So for $w \geq 0$,

$$\begin{aligned} F_W(w) = \mathbb{P}(W \leq w) &= \mathbb{P}(X+Y \leq w) \\ &= \int_{x=0}^{w} \int_{y=0}^{w-x} f_{X,Y}(x,y)\, dy\, dx \\ &= \int_{x=0}^{w} \int_{y=0}^{w-x} e^{-x} e^{-y}\, dy\, dx \\ &= \int_{x=0}^{w} \left[-e^{-x}e^{-y}\right]_{y=0}^{w-x}\, dx \\ &= \int_{x=0}^{w} \left(-e^{-x}e^{-(w-x)} + e^{-x}\right) dx \\ &= \int_{x=0}^{w} e^{-x} - e^{-w}\, dx \\ &= \left[-e^{-x} - xe^{-w}\right]_{x=0}^{w} \\ &= -e^{-w} - we^{-w} + 1 + 0 = 1 - e^{-w} - we^{-w} \end{aligned}$$

That is,

$$F_W(w) = \begin{cases} 0 & w < 0, \\ 1 - e^{-w} - we^{-w} & w \geq 0. \end{cases}$$

Differentiating to obtain density function,

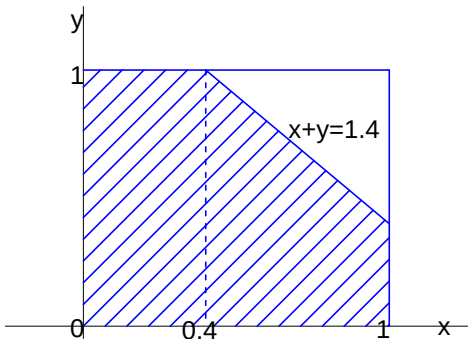
$$\begin{aligned} f_W(w) = \frac{dF_W}{dw} &= \begin{cases} 0 & w < 0, \\ e^{-w} - e^{-w} + we^{-w} & w \geq 0, \end{cases} \\ &= \begin{cases} 0 & w < 0, \\ we^{-w} & w \geq 0. \end{cases} \end{aligned}$$

Consulting the AMS formula sheet, and noting that $\Gamma(2) = (2-1)! = 1$, we see that this matches the pdf of the $\Gamma(2,1)$ distribution, and so $W \sim \Gamma(2,1)$.

2. Since X and Y are independent, their joint density function is

$$\begin{aligned} f_{X,Y}(x,y) &= f_X(x) f_Y(y) \qquad \text{(independence)} \\ &= \begin{cases} 6xy^2 & 0 < x < 1,\ 0 < y < 1, \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

To find $\mathbb{P}(X+Y \leq 1.4)$, integrate over shaded region below:



So now

$$\begin{aligned}
\mathbb{P}(X+Y\le 1.4) &= \int_{x=0}^{0.4}\int_{y=0}^{1} 6xy^2\,dy\,dx + \int_{x=0.4}^{1}\int_{y=0}^{1.4-x} 6xy^2\,dy\,dx \\
&= \int_{x=0}^{0.4}\left[2xy^3\right]_{y=0}^{1} dx + \int_{x=0.4}^{1}\left[2xy^3\right]_{y=0}^{1.4-x} dx \\
&= \int_{x=0}^{0.4} 2x\,dx + \int_{x=0.4}^{1} 2x(1.4-x)^3\,dx \\
&= \left[x^2\right]_{x=0}^{0.4} + \left[\frac{2x(1.4-x)^4}{-4}\right]_{x=0.4}^{1} - \int_{x=0.4}^{1}\frac{2(1.4-x)^4}{-4}\,dx \\
&\quad \text{(using integration by parts for the second integral)} \\
&= 0.4^2 - \frac{0.4^4}{2} + \frac{0.4}{2} + \left[\frac{(1.4-x)^5}{-10}\right]_{x=0.4}^{1} \\
&= 0.16 - 0.0128 + 0.2 - \frac{0.4^5}{10} + \frac{1}{10} \\
&= 0.16 - 0.0128 + 0.2 - 0.001024 + 0.1 = 0.446176
\end{aligned}$$

3. Random variables X and Y are jointly continuous and independent, so have joint pdf $f_{X,Y}(x,y) = f_X(x)f_Y(y)$ (by independence), where $f_X(x)$, $f_Y(y)$ are the marginal pdfs of X, Y, respectively.

   Suppose $y \in \mathbb{R}$ is such that $f_Y(y) > 0$. Then

$$\begin{aligned}
\mathbb{E}(X \mid Y=y) &= \int_{\mathbb{R}} x f_{X|Y}(x \mid y)\,dx \qquad \text{(definition of } E(X \mid Y=y)) \\
&= \int_{\mathbb{R}} x\,\frac{f_{X,Y}(x,y)}{f_Y(y)}\,dx \qquad \text{(definition of } f_{X|Y}(x \mid y)) \\
&= \int_{\mathbb{R}} x\,\frac{f_X(x)f_Y(y)}{f_Y(y)}\,dx \qquad \text{(independence)} \\
&= \int_{\mathbb{R}} x f_X(x)\,dx \\
&= \mathbb{E}(X),
\end{aligned}$$

   as required.

4. (a) Told that X, Y are jointly continuous with joint density $f_{X,Y}(x,y)$. Then

$$\begin{aligned}
\int_{\mathbb{R}} \mathbb{E}(X \mid Y=y) f_Y(y)\,dy &= \int_{y\in\mathbb{R}} \left(\int_{x\in\mathbb{R}} x f_{X|Y}(x \mid y)\,dx\right) f_Y(y)\,dy \\
&\quad \text{(definition of } \mathbb{E}(X \mid Y=y)) \\
&= \int_{y\in\mathbb{R}}\int_{x\in\mathbb{R}} x f_{X|Y}(x \mid y) f_Y(y)\,dx\,dy \\
&= \int_{y\in\mathbb{R}}\int_{x\in\mathbb{R}} x f_{X,Y}(x,y)\,dx\,dy \\
&= \int_{x\in\mathbb{R}} x\left(\int_{y\in\mathbb{R}} f_{X,Y}(x,y)\,dy\right) dx \\
&= \int_{x\in\mathbb{R}} x f_X(x)\,dx \\
&= \mathbb{E}(X),
\end{aligned}$$

   as required.

   (b) We are told that

$$\begin{aligned}
X &= \text{minutes past 8am when alarm rings} \sim \mathrm{U}(0,60) \\
Y &= \text{minutes past 8am when Sam gets up}
\end{aligned}$$

   and that given that $X = x$, Y is uniformly distributed between x and 60.

i. Given that $X = x$, Y is uniformly distributed on the interval $(x, 60)$. So conditional density $f_{Y|X}(y \mid x)$ is

$$f_{Y|X}(y \mid x) = \begin{cases} \dfrac{1}{60 - x} & x < y < 60, \\ 0 & \text{otherwise.} \end{cases}$$

(Note: length of the interval $(x, 60)$ is $60 - x$.)

ii. Given that $X = x$, Y is uniformly distributed on the interval $(x, 60)$, and the expectation of this uniform distribution is the midpoint of the interval $(x, 60)$, so

$$\mathbb{E}(Y \mid X = x) = \frac{x + 60}{2}$$

We are told that $X \sim \mathrm{U}(0, 60)$, so X has pdf

$$f_X(x) = \begin{cases} \dfrac{1}{60} & 0 < x < 60, \\ 0 & \text{otherwise.} \end{cases}$$

Using the Law of Total Expectation,

$$\begin{aligned} \mathbb{E}(Y) &= \int_{x \in \mathbb{R}} \mathbb{E}(Y \mid X = x) f_X(x)\, dx \\ &= \int_{x=0}^{60} \left(\frac{x + 60}{2} \right) \frac{1}{60}\, dx \\ &= \left[\frac{x^2}{240} + \frac{x}{2} \right]_{x=0}^{60} = 15 + 30 = 45 \text{ minutes} \end{aligned}$$

5. (a) For $0 < x < 1$, the conditional distribution of Y given $X = x$ has density $f_{Y|X}(y \mid x) = xe^{-xy}$ for $y > 0$ (and density zero otherwise), which is the density of an exponential distribution with rate parameter x.

Hence, using formula for expectation of an exponential distribution from AMS formula sheet, $\mathbb{E}(Y \mid X = x) = \frac{1}{x}$ for $0 < x < 1$.

(b) Using Law of Total Expectation,

$$\begin{aligned} E(Y) &= \int_{x=0}^{1} \mathbb{E}(Y \mid X = x) f_X(x)\, dx \\ &= \int_{x=0}^{1} \left(\frac{1}{x} \right) 3x^2\, dx = \int_{x=0}^{1} 3x\, dx = \left[\frac{3x^2}{2} \right]_{x=0}^{1} = \frac{3}{2} \end{aligned}$$

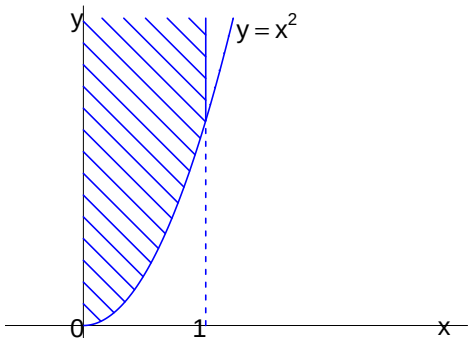
(c) Using Law of Total Probability,

$$\begin{aligned}
\mathbb{P}(Y>3) &= \int_{x=0}^{1} \mathbb{P}(Y>3 \mid X=x) f_X(x)\,dx \\
&= \int_{x=0}^{1} \left(\int_{y=3}^{\infty} f_{Y|X}(y \mid x)\,dy \right) f_X(x)\,dx \\
&= \int_{x=0}^{1} \left(\int_{y=3}^{\infty} xe^{-xy}\,dy \right) 3x^2\,dx \\
&= \int_{x=0}^{1} \left[-e^{-xy} \right]_{y=3}^{\infty} 3x^2\,dx \\
&= \int_{x=0}^{1} 3x^2 e^{-3x}\,dx \\
&= \left[-x^2 e^{-3x} \right]_{x=0}^{1} + \int_{x=0}^{1} 2xe^{-3x}\,dx \qquad \text{(integrating by parts)} \\
&= -e^{-3} + \left[-\frac{2xe^{-3x}}{3} \right]_{x=0}^{1} + \int_{x=0}^{1} \frac{2e^{-3x}}{3}\,dx \qquad \text{(integrating by parts again)} \\
&= -e^{-3} - \frac{2e^{-3}}{3} + \left[-\frac{2e^{-3x}}{9} \right]_{x=0}^{1} \\
&= -\frac{5e^{-3}}{3} - \frac{2e^{-3}}{9} + \frac{2}{9} = \frac{2}{9} - \frac{17e^{-3}}{9} \approx 0.12818
\end{aligned}$$

(d) Joint density is

$$\begin{aligned}
f_{X,Y}(x,y) &= f_X(x) f_{Y|X}(y \mid x) \\
&= \begin{cases} 3x^3 e^{-xy} & 0<x<1,\ 0<y<\infty, \\ 0 & \text{otherwise.} \end{cases}
\end{aligned}$$

To find $\mathbb{P}(Y > X^2)$, integrate over region of non-zero density within which the inequality is satisfied — shaded region below:



$$\begin{aligned}
\mathbb{P}(Y > X^2) &= \int_{x=0}^{1}\int_{y=x^2}^{\infty} 3x^3 e^{-xy}\,dy\,dx \\
&= \int_{x=0}^{1} \left[-3x^2 e^{-xy}\right]_{y=x^2}^{\infty} dx \\
&= \int_{x=0}^{1} 3x^2 e^{-x^3}\,dx \\
&= \int_{x=0}^{1} \frac{d}{dx}\left(-e^{-x^3}\right) dx \qquad \text{(alternatively, use substitution } u = x^3\text{)} \\
&= \left[-e^{-x^3}\right]_{x=0}^{1} = 1 - e^{-1} \approx 0.63212
\end{aligned}$$

6. For two random variables X, Y,

$$\begin{aligned}
\mathbb{V}\text{ar}(X+Y) &= \mathbb{E}\left[(X+Y)^2\right] - (\mathbb{E}(X+Y))^2 \\
&= \mathbb{E}\left[X^2 + 2XY + Y^2\right] - (\mathbb{E}(X) + \mathbb{E}(Y))^2 \\
&= \mathbb{E}(X^2) + 2\mathbb{E}(XY) + \mathbb{E}(Y^2) - (\mathbb{E}(X))^2 - 2\mathbb{E}(X)\mathbb{E}(Y) - (\mathbb{E}(Y))^2 \\
&= \left(\mathbb{E}(X^2) - (\mathbb{E}(X))^2\right) + 2\left(\mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)\right) + \left(\mathbb{E}(Y^2) - (\mathbb{E}(Y))^2\right) \\
&= \mathbb{V}\text{ar}(X) + 2\mathbb{C}\text{ov}(X,Y) + \mathbb{V}\text{ar}(Y)
\end{aligned}$$

as required.

7. (a) For random variables X, Y and constants a, b, c, d,

$$\begin{aligned}
\mathbb{C}\text{ov}(a+bX, c+dY) &= \mathbb{E}\left[(a+bX)(c+dY)\right] - \mathbb{E}(a+bX)E(c+dY) \\
&= \mathbb{E}\left[ac + bcX + adY + bdXY\right] - \left(a + b\mathbb{E}(X)\right)\left(c + d\mathbb{E}(Y)\right) \\
&= ac + bc\mathbb{E}(X) + ad\mathbb{E}(Y) + bd\mathbb{E}(XY) \\
&\quad - ac - bc\mathbb{E}(X) - ad\mathbb{E}(Y) - bd\mathbb{E}(X)\mathbb{E}(Y) \\
&= bd\left(\mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)\right) \\
&= bd\,\mathbb{C}\text{ov}(X,Y)
\end{aligned}$$

as required.

(b) For random variables X, Y, Z,

$$\begin{aligned}
\mathbb{C}\text{ov}(X+Y, Z) &= \mathbb{E}\left[(X+Y)Z\right] - \mathbb{E}[X+Y]\mathbb{E}[Z] \\
&= \mathbb{E}\left[XZ + YZ\right] - \left(\mathbb{E}[X] + \mathbb{E}[Y]\right)\mathbb{E}[Z] \\
&= \mathbb{E}[XZ] + \mathbb{E}[YZ] - \left(\mathbb{E}[X]\mathbb{E}[Z] + \mathbb{E}[Y]\mathbb{E}[Z]\right) \\
&= \left(\mathbb{E}[XZ] - \mathbb{E}[X]\mathbb{E}[Z]\right) + \left(\mathbb{E}[YZ] - \mathbb{E}[Y]\mathbb{E}[Z]\right) \\
&= \mathbb{C}\text{ov}(X,Z) + \mathbb{C}\text{ov}(Y,Z)
\end{aligned}$$

as required.

Exam Style Question

(a) The marginal density f_Y is found as follows. For $y > 0$:

$$\begin{aligned}
f_Y(y) &= \int_0^{\infty} \frac{1}{y} e^{-\left(y + \frac{x}{y}\right)}\,\mathrm{d}x \\
&= \left[\frac{1}{y} e^{-y} \frac{e^{-\frac{x}{y}}}{-1/y}\right]_0^{\infty} \\
&= e^{-y}
\end{aligned}$$

For $y < 0$ we have $f_Y(y) = 0$.

That is, Y is exponentially distributed with rate parameter 1, $Y \sim \text{Exp}(1)$.
(3 marks)

(b) (i) The conditional density of X given $Y = y$ for $y > 0$ is given by

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{\frac{1}{y}e^{-\left(y+\frac{x}{y}\right)}}{e^{-y}} = \frac{1}{y}e^{-\frac{x}{y}}$$

for $x > 0$, with $f_{X|Y}(x|y) = 0$ for $x \leq 0$. (2 marks)

(ii) The density $f_{X|Y}(x|y)$ is the density of an exponential random variable with parameter $1/y$, hence

$$\mathbb{E}[X|Y = y] = \left(\frac{1}{y}\right)^{-1} = y$$

as required. (1 mark).

(iii)

$$\mathbb{E}[e^{X/3}|Y = 1] = \int_{x=0}^{\infty} e^{x/3} f_{X|Y}(x|1)\,\mathrm{d}x = \int_{x=0}^{\infty} e^{x/3}e^{-x}\,\mathrm{d}x = \int_{x=0}^{\infty} e^{-\frac{2}{3}x}\,\mathrm{d}x = \frac{3}{2}.$$

(3 marks)

(c) First we find $\mathbb{E}[XY]$:

$$\begin{aligned}\mathbb{E}[XY] &= \int_{y=0}^{\infty}\int_{x=0}^{\infty} xy\frac{1}{y}e^{-\left(y+\frac{x}{y}\right)}\,\mathrm{d}x\,\mathrm{d}y \\ &= \int_{y=0}^{\infty} ye^{-y}\int_{x=0}^{\infty}\frac{x}{y}e^{-\frac{x}{y}}\,\mathrm{d}x\,\mathrm{d}y\end{aligned}$$

As we have seen earlier in the question

$$\int_{x=0}^{\infty}\frac{x}{y}e^{-\frac{x}{y}}\,\mathrm{d}x = \mathbb{E}[X|Y = y] = y.$$

Therefore

$$\mathbb{E}[XY] = \int_{y=0}^{\infty} y^2e^{-y}\,\mathrm{d}y.$$

As we saw in part (a), $Y \sim \mathrm{Exp}(1)$ and has density e^{-y} so we have

$$\mathbb{E}[XY] = \int_{y=0}^{\infty} y^2e^{-y}\,\mathrm{d}y = \mathbb{E}[Y^2].$$

We can find the value of the second moment from the mean and variance as

$$\mathbb{E}[Y^2] = \mathbb{V}\mathrm{ar}(Y) + (\mathbb{E}[Y])^2 = 1 + 1^2 = 2.$$

Therefore

$$\mathbb{C}\mathrm{ov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 2 - 1 \times 1 = 1.$$

(2 marks)

(d) Since $\mathbb{C}\mathrm{ov}(X,Y) \neq 0$ we must have that X and Y are not independent. In fact, $\mathbb{C}\mathrm{ov}(X,Y) > 0$ so there is a positive relationship between X and Y. (1 mark)